

Introduction: working draft

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Grain-boundary oxidation in 304 stainless steel (304SS) exposed to pressurized water reactor (PWR) primary water depends on the chemical changes introduced by prior irradiation. Deng and co-workers found that increasing the proton irradiation dose produced deeper intergranular oxidation in 304 nuclear-grade stainless steel after water exposure, accompanied by a decrease in the Cr content of the intergranular oxide (Deng, Peng, Han, et al. 2017). These observations connect irradiation history to both oxide chemistry and penetration depth. Establishing a quantitative relationship between them requires a description of how the changing oxide composition affects oxygen transport and, consequently, the advance of the oxidation front.

Radiation-induced segregation (RIS) establishes the local alloy composition from which the intergranular oxide develops. Coupling between solute and irradiation-generated point-defect fluxes redistributes alloying elements near grain boundaries. Measurements on irradiated 304 nuclear-grade stainless steel reveal Cr depletion together with Ni and Si enrichment (Deng, Peng, Han, et al. 2017; Deng, Peng, and Han 2021). Subsequent oxidation selectively consumes alloying elements, changing both the oxide composition and the adjacent metal, where Ni enrichment has also been observed (Deng, Peng, and Han 2021). The lower Cr content of the oxide formed on irradiated boundaries has been associated with reduced resistance to oxygen diffusion (Deng, Peng, Han, et al. 2017). Oxide protectiveness therefore has a direct transport meaning: an oxide that restricts oxygen diffusion more strongly limits the oxygen available for further oxidation.

Si oxidation and dissolution also alter this transport resistance by changing the structure through which oxygen moves. Using model alloys with systematically varied Cr and Si contents, Du and co-workers linked increased Si content to oxide degradation and greater cracking susceptibility in hydrogenated water. They attributed this behavior to dissolution of Si-containing oxide and the resulting porous structure, which facilitates oxygen transport (Du et al. 2026). Related microscopy on proton-irradiated 316L identified Si enrichment at oxide tips and porosity behind them, consistent with preferential Si oxidation followed by dissolution (Kuang et al. 2022). These studies provide a mechanistic basis for including dissolution-generated free volume in the transport description. Together with the dependence on Cr content, they motivate representing the evolving oxide composition and free volume through an effective oxygen diffusivity. This connects the segregation of multiple elements to a common quantity governing oxygen transport.

Models coupling solute and point-defect transport can predict the composition redistribution produced by irradiation. For example, Kadambi and co-workers developed a model for segregation at grain boundaries and dislocation structures in austenitic Fe–Cr–Ni (Kadambi et al. 2025). Connecting such predictions to oxidation kinetics requires tracking how the segregated alloy supplies the interfacial reactions, how those reactions change the oxide, and how the resulting diffusivity affects subsequent oxygen delivery. The quantity to be predicted is thus the oxidation response arising from the evolving chemical state. Segregation profiles, oxide composition, interfacial Ni enrichment, and penetration depth provide complementary observables for assessing this coupling.

Here, we develop a coupled RIS and grain-boundary oxidation model for 304SS, represented by the Fe–Cr–Ni–Si system. The model links defect-mediated solute redistribution and selective interfacial consumption to an effective oxygen diffusivity, D_{eff} , determined by the evolving oxide composition and free volume. Within this description, increasing the Cr content of the oxide lowers the effective diffusivity, whereas Si oxidation followed by complete dissolution contributes free volume that enhances transport. The same D_{eff} enters both oxygen transport along the grain boundary and oxygen delivery across the oxide to the metal interface. The model therefore connects RIS to oxidation kinetics through the resulting change in oxygen transport, with metal consumption feeding back on the local composition. The complete-dissolution treatment is a simplification motivated by the mechanism discussed by Du and co-workers (Du et al. 2026). We examine the dose and exposure-time dependence of oxide penetration under sequential irradiation and water exposure, using separate calibration of the RIS and oxidation components and a common set of oxidation parameters across doses. The framework also permits predictions of concurrent RIS and oxidation within the same transport description.

References

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